

Internship Project

Churn Analysis and Customer Intelligence



Internship ID: IBMUEDA7363

Internship Title: IBM SkillsBuild Data Analytics with AI Internship 2026

Project Report By BOBBY AIML Engineer (Final year Student)

CT Institute of Engineering Management and Technology Shahpur Campus

You Learn – Churn Analysis & Customer Intelligence



1. The Business Challenge

In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You will act as a Data Analyst tasked with identifying high-risk subscribers using a multi-dimensional dataset (Customer demographics, Subscription tiers, and Support escalations).

2. Core Tech Stack

SQL & Python Integration (*numpy, pandas, sqlite3, matplotlib, seaborn*), Performance Engineering (*data cleaning, feature engineering, analytics*), Behavioral Visualization, Writing Actionable Insights

3. Project Milestones

- Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets.
- Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.
- Executive Reporting: Translating technical findings into billion-dollar business insights.

Churn Analysis - Roadmap

- Connect SQL database to Python – *pandas and sqlite3*
- Data import using SQL query in Python – *pandas and sqlite3*
- Data Cleaning – *numpy and pandas*
- data types, rename cols, select specific cols, QCs, handle missing/null values
- Feature Engineering – *numpy and pandas*
- Create new calculated cols, data transformation, use filters
- Data Analysis – *numpy and pandas*
- EDA - aggregation, group by, pivot table
- Data Visualization – *matplotlib and seaborn*

Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you.

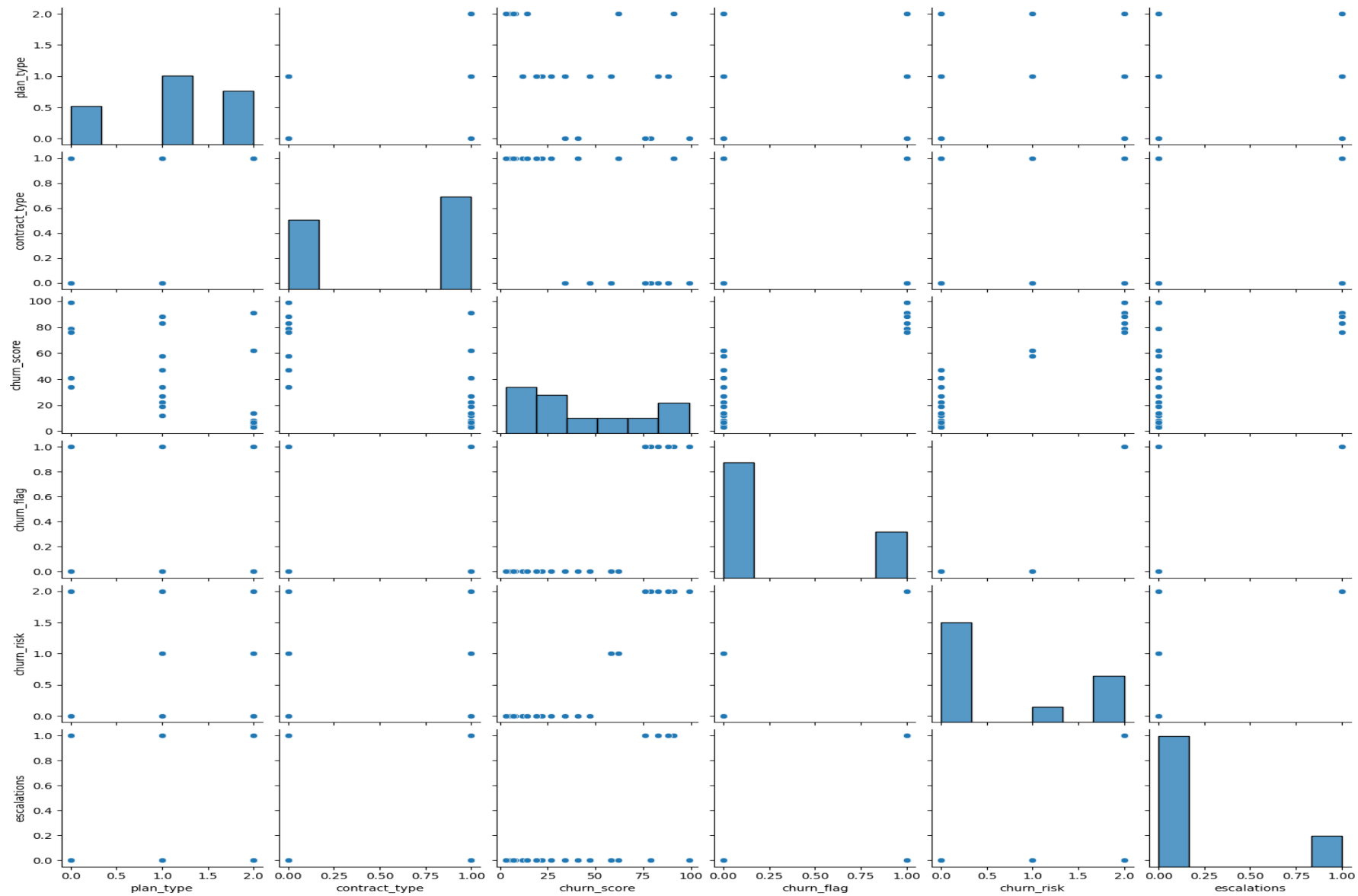
For this, we only need to look at three things:

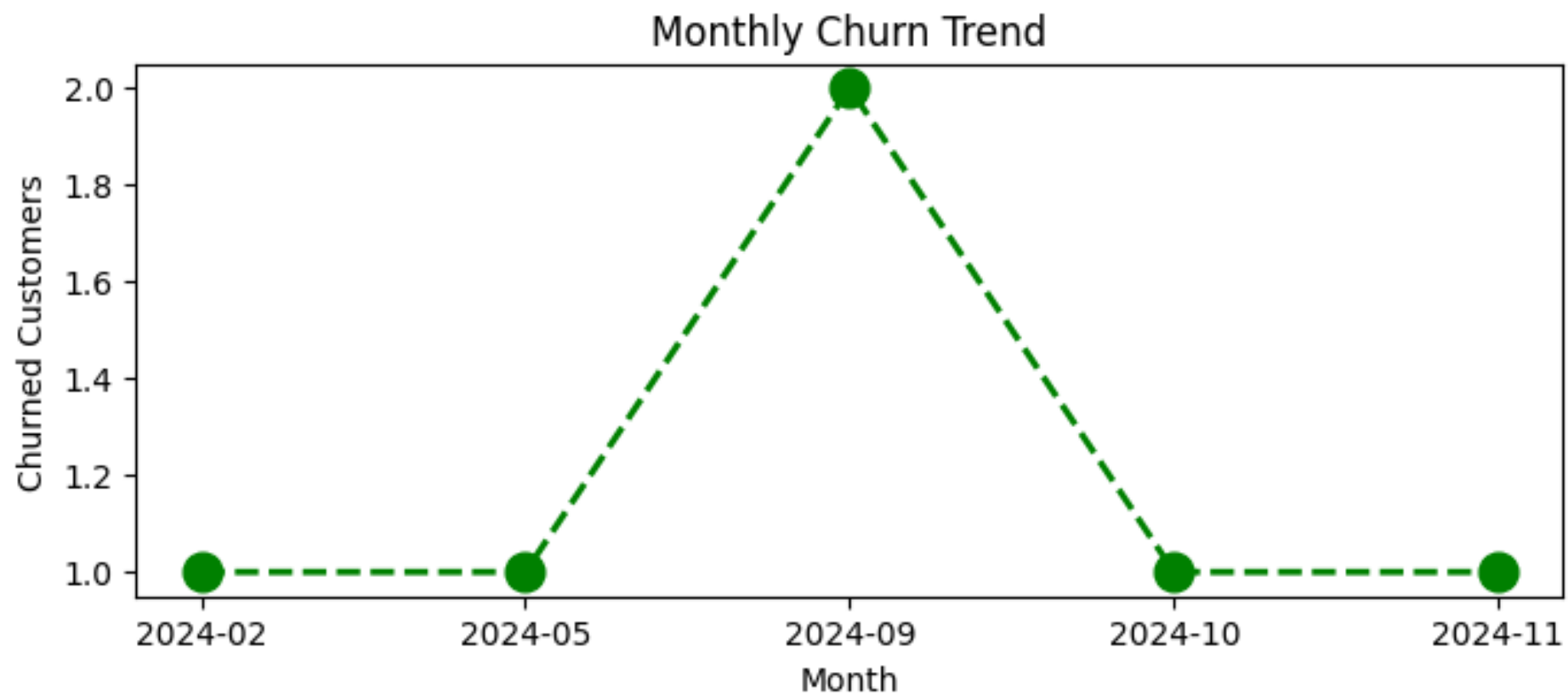
- **"Who"**: Identifying which customers left or will leave in future
- **"Why"**: Analyzing their behavior before they left.
- **"When"**: Finding the "danger zone" – point when a user leave

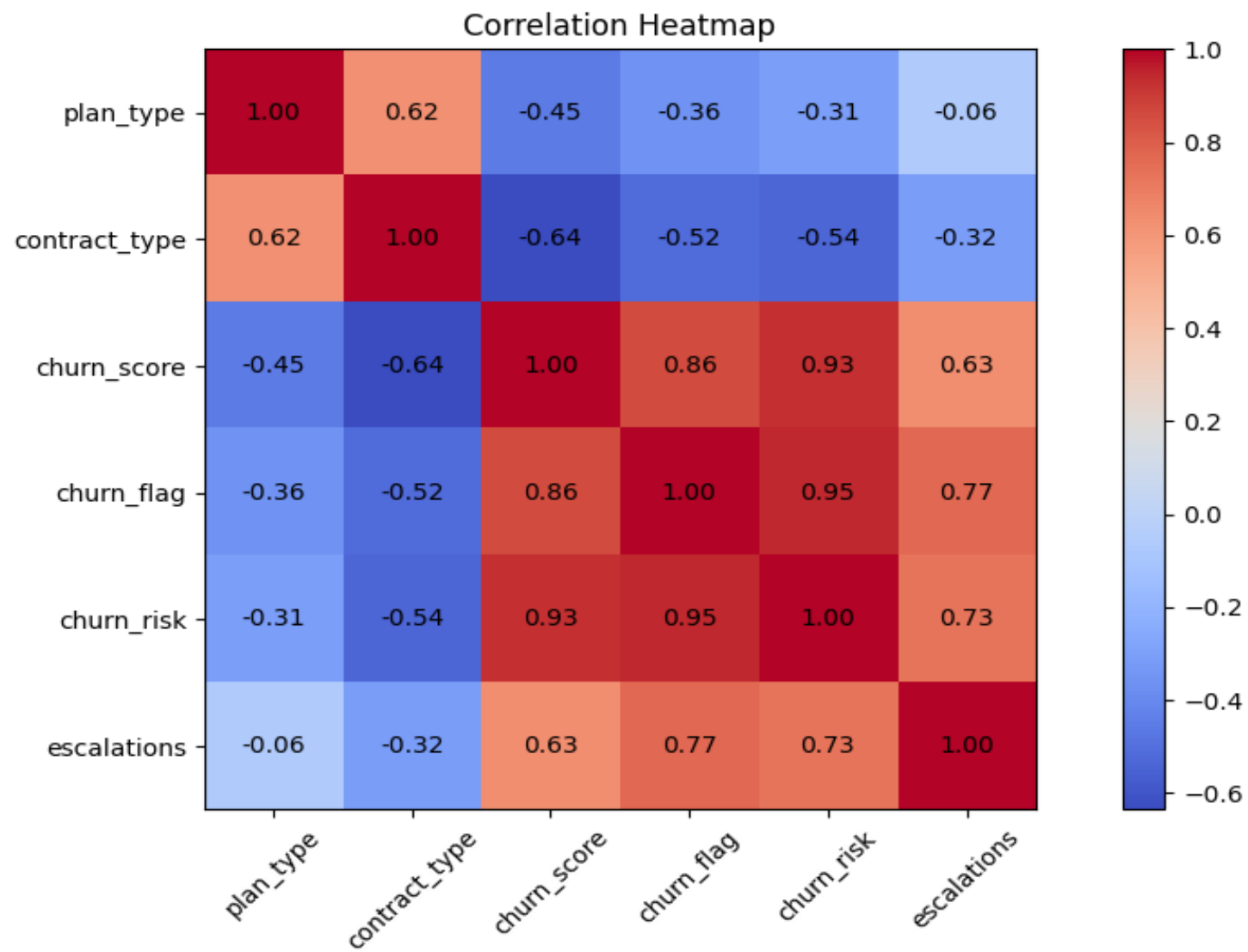
Business Type	Churn Definition
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Using service/app in 90 – 120 days
Streaming	Membership inactive
Telecom	Account terminated

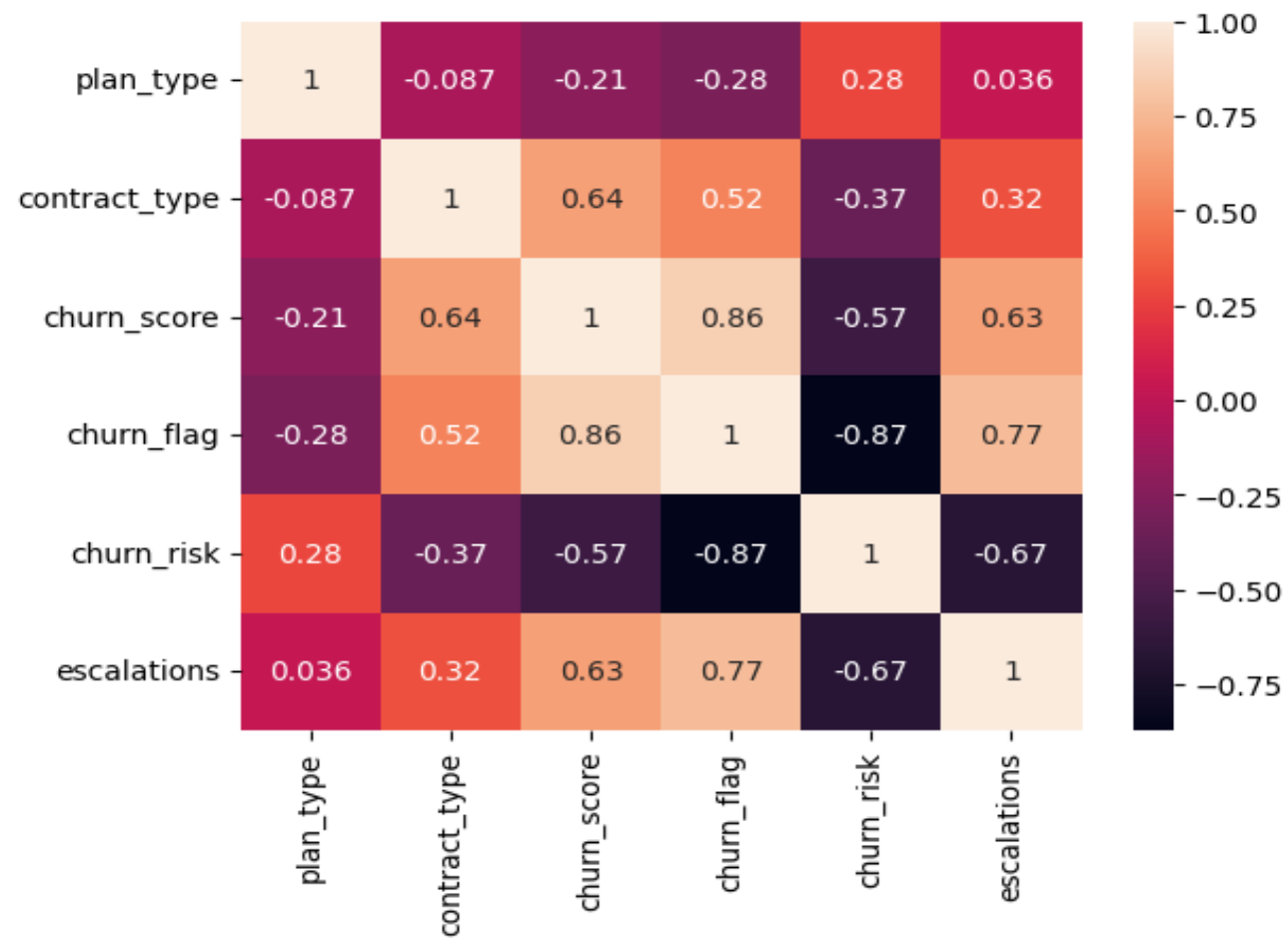
4. Database: customer_churn

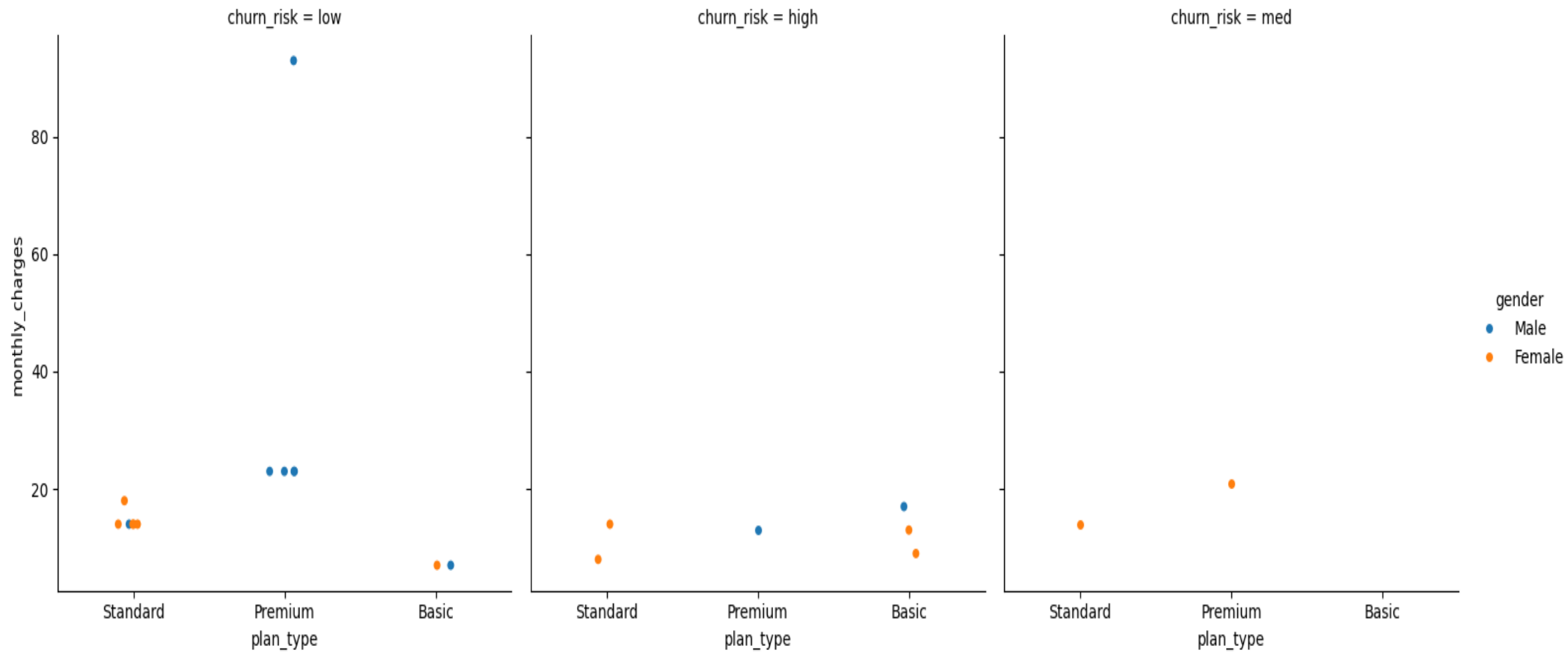
db_customer	db_subscription	db_support
customerid	customerid	customerid
name	subscription_start_date	complaint_date
country	subscription_type	Escalations
State	renewal_date	csat_score
gender	plan_type	col_1
dob	contract_type	comment
interests	cancellation_date	
pincode	cancellation_reason	
	monthly_charges	
	cltv	
	churn_score	

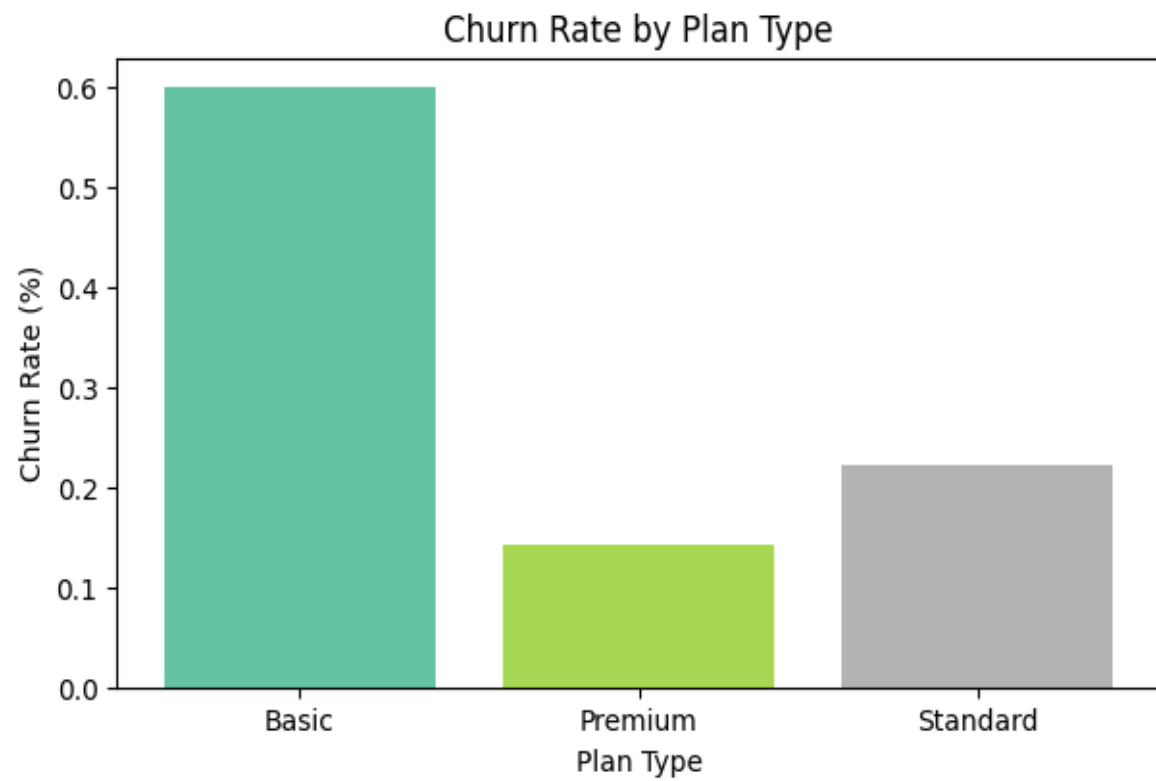




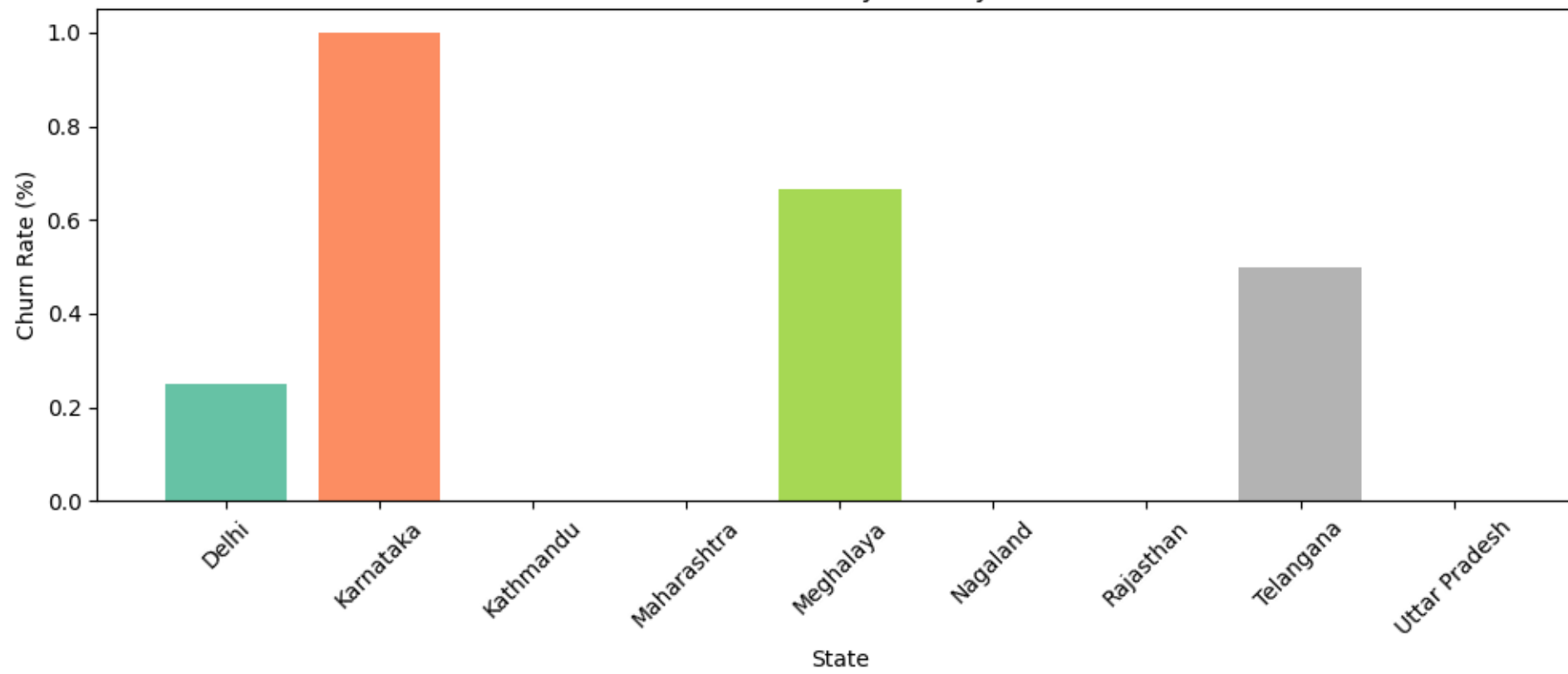








Churn Rate by Country



Calculated metrics

KPI	Formula
churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1-churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100
Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

Insights

Insights

- Churn Rate: 28.6% | Retention Rate: 71.4%
- Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact
- Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka
- Average Tenure (Days): 1,451 | ARPU is Rs 18.8
- Total Revenue = 395
- Revenue loss due to churn = 74 | CLTV Lost = 2,047
- % Revenue loss = 18%
- monthly vs annual churn = 55.6% vs 8.3%

Action Items

- Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc
- Did we increased subs price for **Basic plan** or recently made any changes – specially in sep month?
- Check what competitor is doing – bcoz one user has moved to competitor
- Focus on customers with 'High' & 'Med' churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, sms, calls and fix their issue.

